

Saish Karole

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PROFESSIONAL SUMMARY

I am an avid learner who has completed projects in Computer Architecture and FPGA programming. I actively seek opportunities in Computer Architecture, RTL Design, Verification, and Embedded Systems.

SKILLS

Languages: C, C++, Embedded C, Python, Verilog HDL

HDL Tools: Yosys, Icarus Verilog, GTKWave, Quartus Prime, Vivado, Modelsim, Vitis IDE

Embedded Platforms: ESP-IDF, FreeRTOS, Raspberry Pi, Arduino, KV260

Developer Tools: Git/Github, Docker, CMake, Makefile, VS Code

Others: Linux, OpenCV, ScalaTest, Chisel HDL

EXPERIENCE

Design Intern

May 2025 – Jul 2026

Aumnatic Systems Pvt Ltd

Turbhe, Navi Mumbai

- Developed firmware for an ESP32-based prototype to test DC brushed motors under load.
- Implemented real-time voltage control, stall current monitoring, and data logging.
- Created a Python WebSocket client for remote control and real-time data visualization.

Team Member

Aug. 2023 – Mar. 2024

e-Yantra Robotics Competition (eYRC), IIT Bombay - Semi-finalists

Github

- Built a Pick and Place + Line Following robot simulating a space station task.
- Designed a single-cycle RISC-V RV32I core in Verilog to execute compiled C code.
- Ran Dijkstra's Algorithm on FPGA to compute shortest paths using a PID-controlled robot.
- Programmed on DE0-Nano (Cyclone IV FPGA) using Intel Quartus and Modelsim.

PROJECTS

RISC-V RV32IM CPU on FPGA

Aug. 2023 – Oct. 2023

Github

- Designed and simulated a single-cycle RISC-V core in Verilog with testbenches.
- Deployed on UPduino 3.0 using Yosys + IceStorm; verified Fibonacci output on 7-segment display.

Chisel Digital Designs

Jun. 2024 – Jul. 2024

Github

- Learned and applied Chisel HDL to build and simulate basic circuits using ScalaTest.
- Developed adders, encoders, and simple FSMs.

Matrix Multiplication on FPGA

Dec. 2024 – Feb. 2025

Github

- Implemented matrix multiplication in Verilog with a C kernel for acceleration on KV260.
- Validated results via Vitis IDE and performance testing on Kria board.

EDUCATION

Veermata Jijabai Technological Institute (VJTI)

Mumbai, India

B.Tech in Electronics Engineering | CGPA: 8.32 (Till Sem 6)

Nov. 2022 – Present

- Relevant Coursework: Digital Electronics, Microprocessors and Microcontrollers, Signals and Systems, Instrumentation, Engineering Math.

EXTRA-CURRICULAR ACTIVITIES

Society of Robotics and Automation (SRA), VJTI

Aug. 2023 – Present

Active Member

Website

- Organized robotics and embedded systems workshops for 220+ students.
- Conducted "Wall-E", "Pixels", and "MARIO" workshops on line following and self-balancing bots, computer vision, and micro-ROS.
- Mentored two FPGA-based project teams on Arty A7-35T under Eklavya Mentorship.